Computer Networks Lab

[ENCS352]

KR. MANGALAM UNIVERSITY

For

BACHELORS IN TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING

With Specialisation In AI & ML

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EXPERIMENT 1

(a): CPT PLATFORM OVERVIEW

Theory

Cisco Packet Tracer is a network simulation tool used to design and test network topologies without physical hardware. It provides a virtual environment with devices like routers, switches, and PCs. It helps students understand networking concepts. It supports both real-time and simulation modes. It is widely used for learning and experimentation.

Steps

1. Open Cisco Packet Tracer.
2. Observe workspace, device toolbar, and connections panel.
3. Drag one PC and one switch.
4. Connect them using a straight-through cable.
5. Click on devices to explore Desktop and Config tabs.

Result

Successfully explored the Cisco Packet Tracer interface.

Conclusion

Packet Tracer provides an effective platform for learning networking concepts.

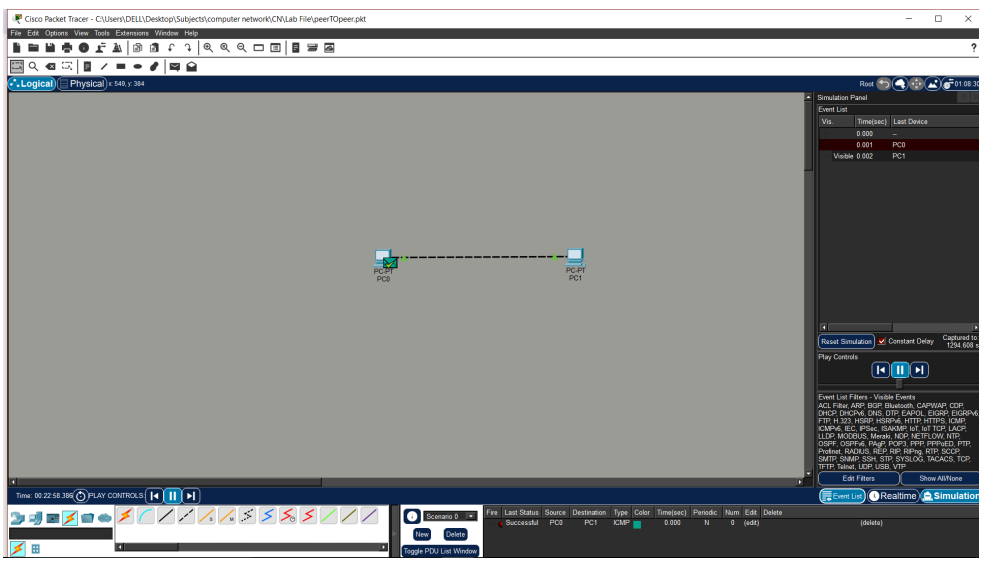
(b): PEER-TO-PEER NETWORK

Theory

A peer-to-peer network allows devices to communicate directly without a central server. Each device acts as both client and server. It is simple, cost-effective, and suitable for small networks. Data sharing can be done directly between devices. However, it lacks centralized control and security.

Steps

1. Add two PCs (PC0, PC1).
2. Connect using a cross-over cable.
3. Assign IP addresses:
    a. PC0 → 192.168.1.1
    b. PC1 → 192.168.1.2
4. Open Command Prompt on PC0.
5. Run: ping 192.168.1.2



Result

Ping successful between PCs. Peer-to-peer communication was established successfully.

EXPERIMENT 2

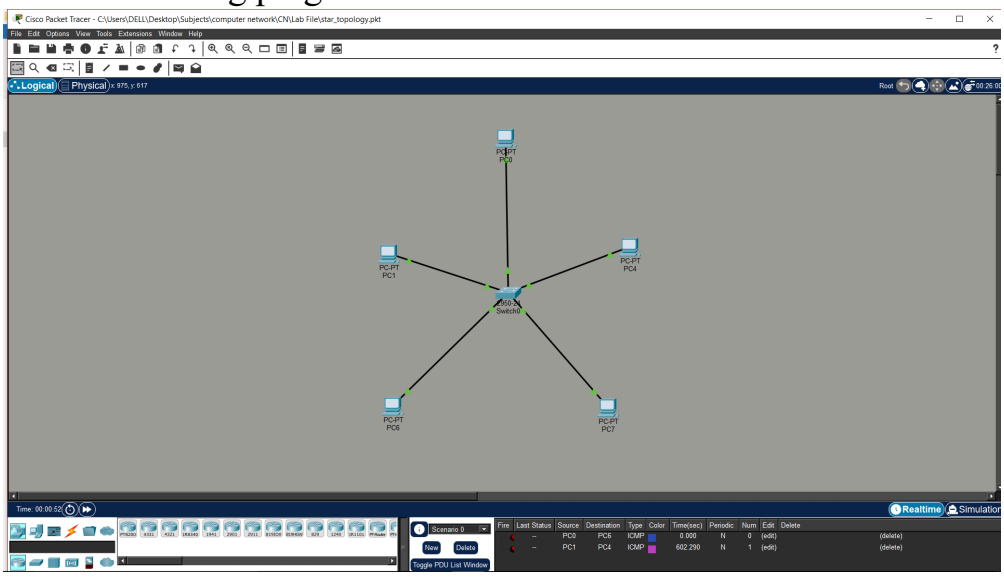
(a): STAR TOPOLOGY

Theory

Star topology connects all devices to a central switch. All communication passes through this central device. It is easy to install and manage. Failure of one link does not affect others. However, failure of the central device stops the entire network.

Steps

1. Add one switch and multiple PCs.
2. Connect PCs to switch using straight cable.
3. Assign IPs:
    a. PC0 → 192.168.1.1
    b. PC1 → 192.168.1.2
    c. PC2 → 192.168.1.3
    d. PC3 → 192.168.1.4
4. Test using ping.



Result

All devices communicate through switches. Star topology is reliable and easy to troubleshoot.

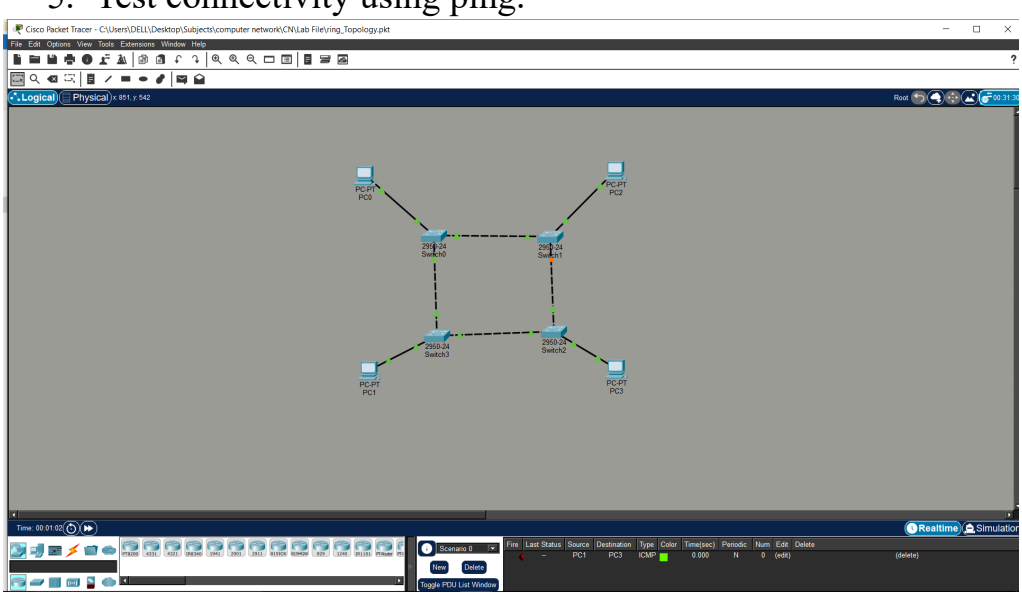
(b): RING TOPOLOGY

Theory

Ring topology connects devices in a circular form where each node is connected to two others. Data travels in one direction around the ring. It ensures equal access to all nodes. However, failure of a single link can disrupt the entire network. It is less commonly used in modern networks.

Steps

1. Add 4 PCs and 4 switches.
2. Connect switches in circular form.
3. Connect PCs to switches.
4. Assign IPs:
    a. PC0 → 192.168.1.1
    b. PC1 → 192.168.1.2
    c. PC2 → 192.168.1.3
    d. PC3 → 192.168.1.4
5. Test connectivity using ping.



Result

Communication is successful in ring structure. Ring topology works but is less reliable than stars.

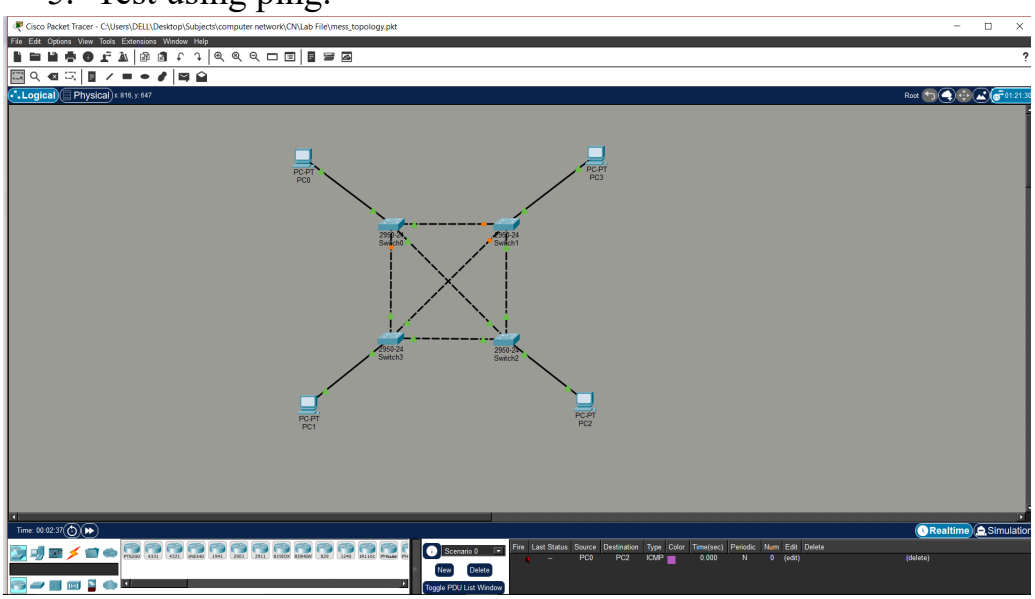
(c): MESH TOPOLOGY

Theory

Mesh topology connects every device to every other device. It provides multiple paths for data transmission. This increases reliability and fault tolerance. Even if one path fails, communication continues. However, it is expensive and complex to implement.

Steps

1. Add switches and PCs.
2. Connect each switch with every other switch.
3. Connect PCs to switches.
4. Assign IPs:
    a. PC0 → 192.168.1.1
    b. PC1 → 192.168.1.2
    c. PC2 → 192.168.1.3
    d. PC3 → 192.168.1.4
5. Test using ping.



Result

Multiple communication paths are established. Mesh topology provides high reliability.

(d): TREE TOPOLOGY

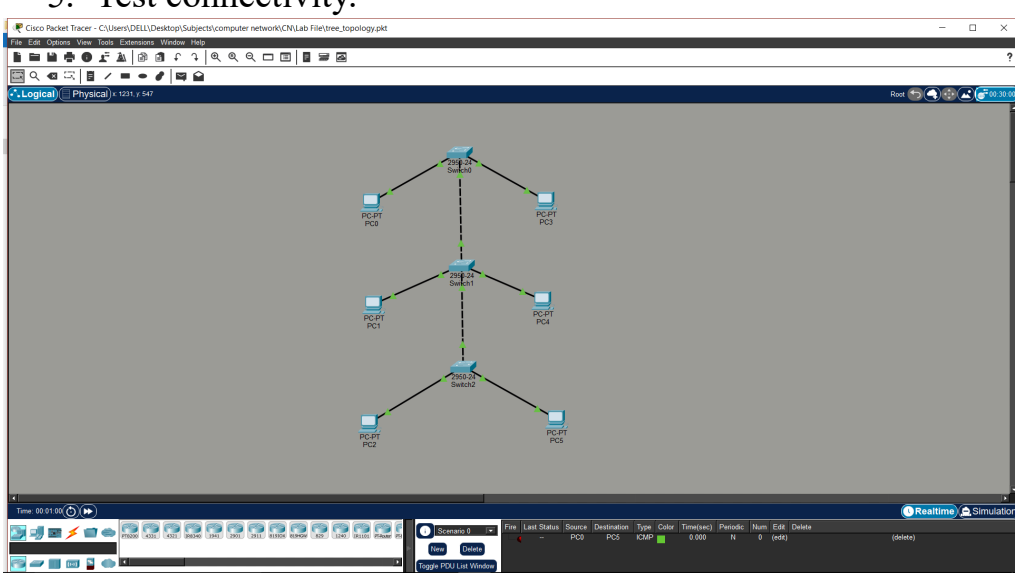
Theory

Tree topology is a hierarchical network structure combining multiple star topologies. It has a root node and multiple levels. It is suitable for large networks. It allows for easy expansion and management. However, failure of backbone can affect the entire network.

Steps

1. Add multiple switches in hierarchy.
2. Connect switches in parent-child structure.
3. Connect PCs to switches.
4. Assign IPs:
    a. PC0 → 192.168.1.1
    b. PC1 → 192.168.1.2
    c. PC2 → 192.168.1.3
    d. PC3 → 192.168.1.4
    e. PC4 → 192.168.1.5
    f. PC5 → 192.168.1.6

5. Test connectivity.



Result

Hierarchical communication was established. Tree topology is scalable and efficient.

EXPERIMENT 3

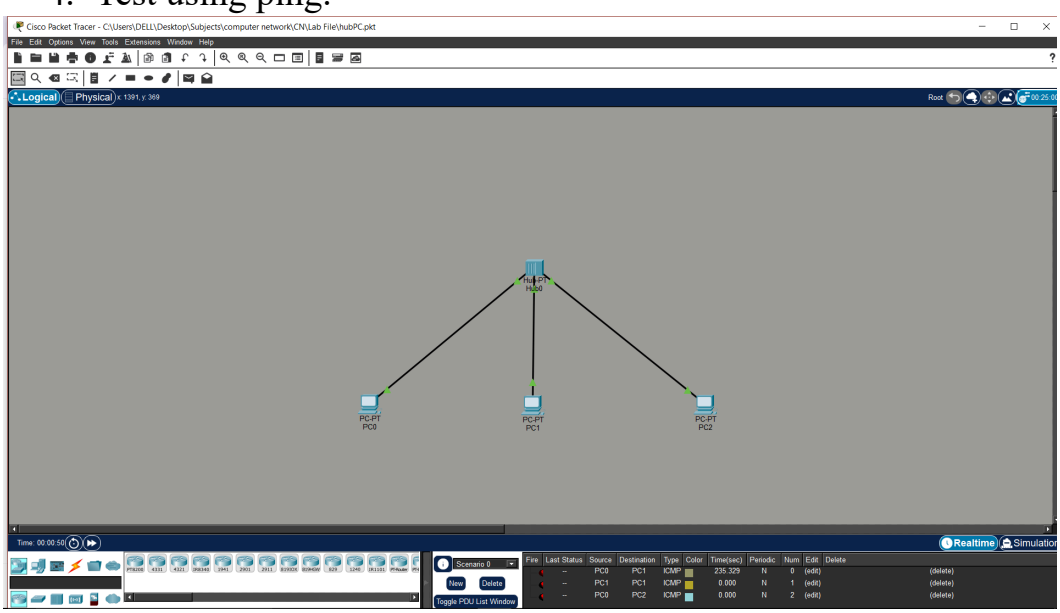
(a): LAN USING HUB

Theory

A hub is a basic networking device that broadcasts data to all connected devices. It operates on the physical layer. It does not filter data, leading to collisions. It is simple but inefficient. It is mostly replaced by switches.

Steps

1. Add one hub and three PCs.
2. Connect PCs to hub.
3. Assign IPs:
    a. PC0 → 192.168.1.1
    b. PC1 → 192.168.1.2
    c. PC2 → 192.168.1.3
4. Test using ping.



Result

All devices receive transmitted data. Hub-based LAN is inefficient due to broadcasting.

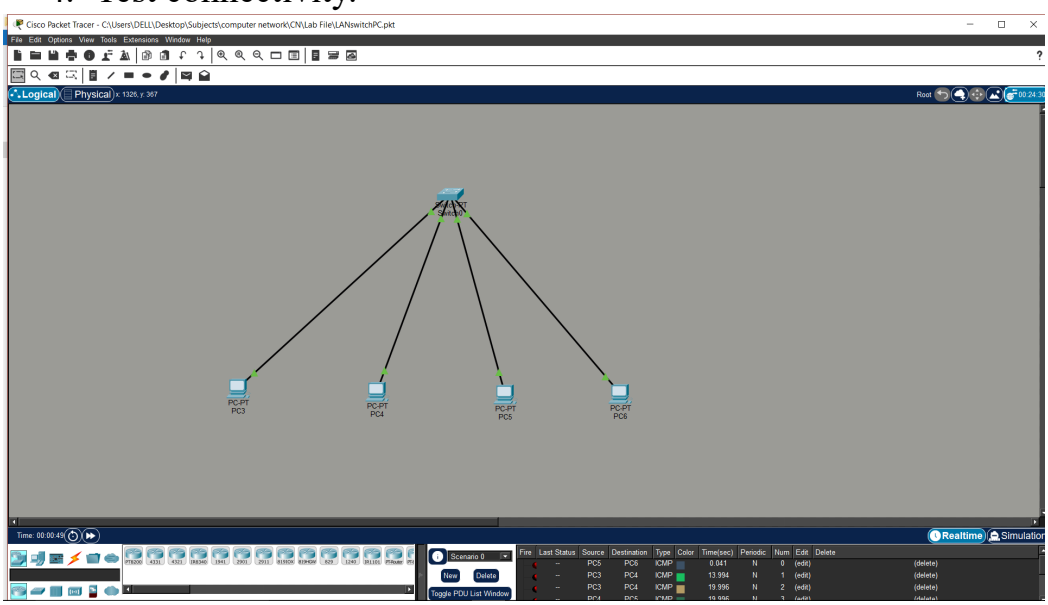
(b): LAN USING SWITCH

Theory

A switch is a networking device that forwards data only to the destination device. It operates on the data link layer. It reduces collisions and improves performance. It uses MAC address tables for efficient communication. It is widely used in LANs.

Steps

1. Add one switch and multiple PCs.
2. Connect PCs to switch.
3. Assign IPs:
    a. PC3 → 192.168.1.1
    b. PC4 → 192.168.1.2
    c. PC5 → 192.168.1.3
    d. PC6 → 192.168.1.4
4. Test connectivity.



Result

Efficient communication is achieved. Switches improve network performance.

EXPERIMENT 4

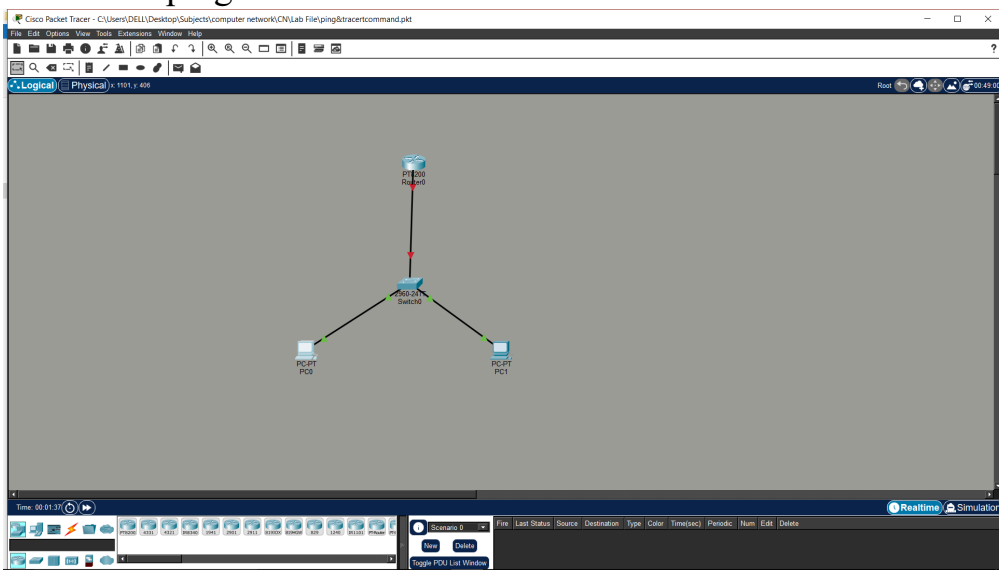
PING & TRACERT

Theory

Ping is used to test connectivity between devices using the ICMP protocol. Tracert shows the path taken by packets. These tools help troubleshoot network issues. They provide information about delays and failures. They are essential diagnostic tools.

Steps

1. Add router, switch, and PCs.
2. Connect devices properly.
3. Assign IPs:
    a. PC0 → 192.168.1.1
    b. PC1 → 192.168.1.2
    c. Router → 192.168.1.254
4. Set gateway in PCs → 192.168.1.254
5. Run ping and tracert commands.



Result

Connectivity and path verified successfully. Ping and tracert help diagnose network problems.

EXPERIMENT 5

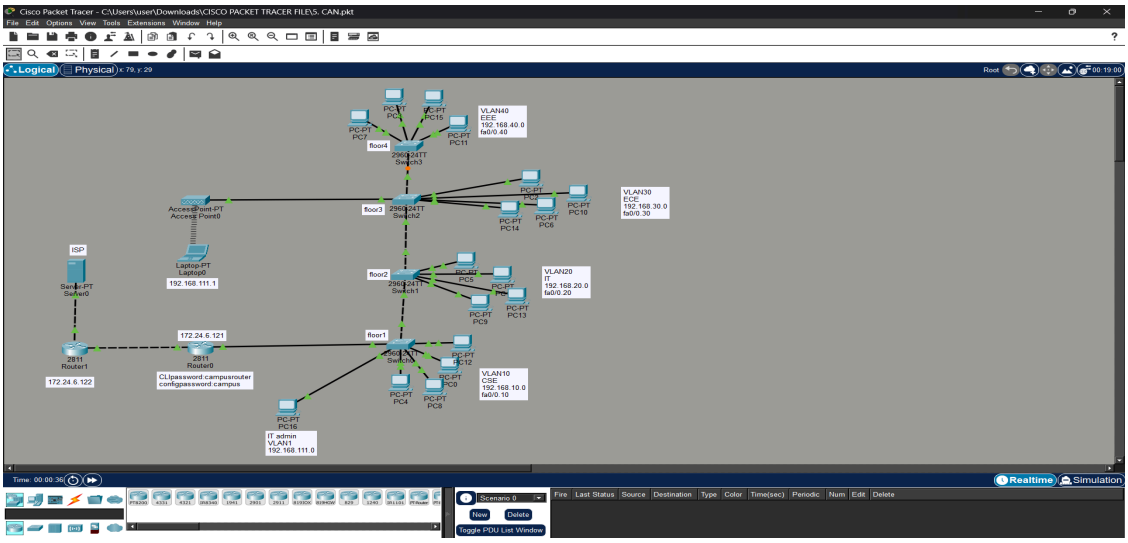
CAMPUS AREA NETWORK (CAN)

Theory

A Campus Area Network connects multiple LANs across buildings. It includes routers, switches, and servers. It allows communication between departments.

Steps

1. Add router, multilayer switch, switches, PCs, servers.
2. Divide network into departments.
3. Connect all switches to central switch.
4. Assign IPs:
    a. ADMIN → 192.168.1.1
    b. HR → 192.168.2.1
    c. FINANCE → 192.168.3.1
    d. BUSINESS → 192.168.4.1
    e. E&C → 192.168.5.1
    f. A&D → 192.168.6.1
    g. STUDENT LAB → 192.168.7.1
5. Set gateways → 192.168.X.254
6. Test communication.



Result - All departments connected successfully. CAN enables efficient campus-wide communication.

EXPERIMENT 6

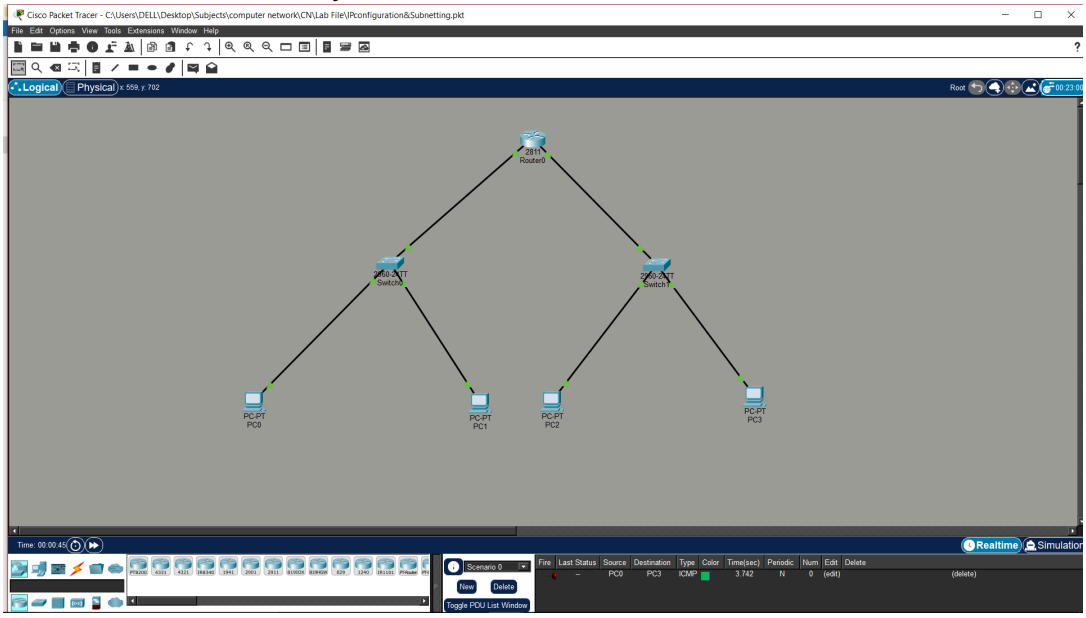
IP CONFIGURATION & SUBNETTING

Theory

IP addressing uniquely identifies devices in a network. Subnetting divides a large network into smaller subnetworks. It improves performance and security. Each subnet has its own range. It reduces congestion and improves management.

Steps

1. Add router, switches, and PCs.
2. Divide network:
    a. Subnet1 → 192.168.1.0/25
    b. Subnet2 → 192.168.1.128/25
3. Assign IPs accordingly.
4. Configure router interfaces.
5. Set the gateways.
6. Test connectivity.



Result

Subnetted network works correctly. Subnetting improves efficiency and organization.

EXPERIMENT 7

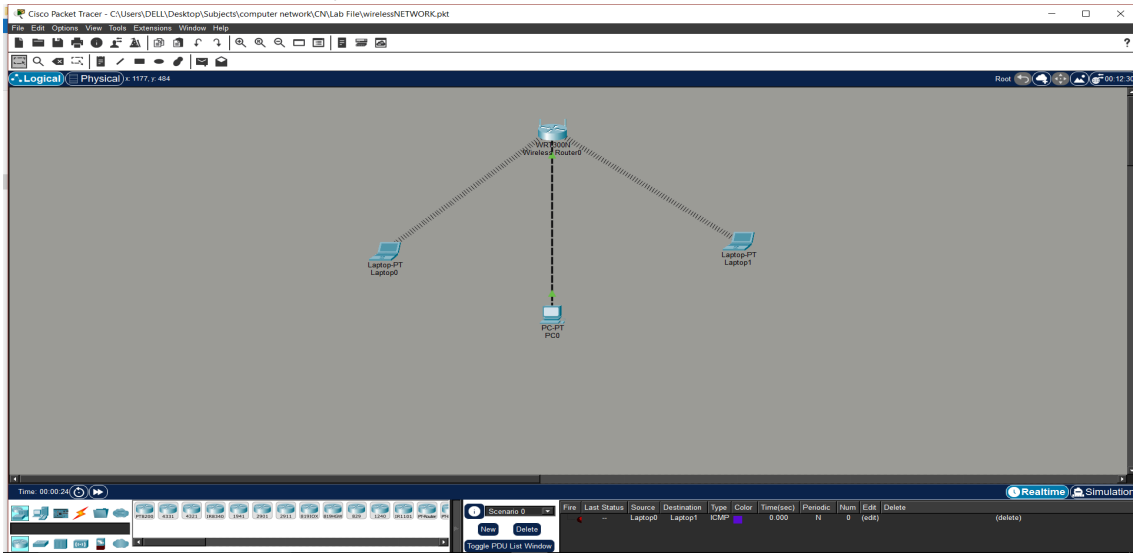
WIRELESS NETWORK

Theory

Wireless networks connect devices without cables using radio signals. A wireless router provides connectivity through SSID. Devices like laptops connect using Wi-Fi. It provides flexibility and mobility. It is widely used in homes and offices.

Steps

1. Add wireless router and laptops.
2. Configure SSID and password.
3. Connect laptops wirelessly.
4. Assign IPs:
    a. Router → 192.168.0.1
    b. Laptop0 → 192.168.0.2
    c. Laptop1 → 192.168.0.3
5. Test connectivity.



Result

Wireless communication is established. Wireless networks provide flexibility and ease of use.

EXPERIMENT 8

CAPSTONE PROJECT

Theory

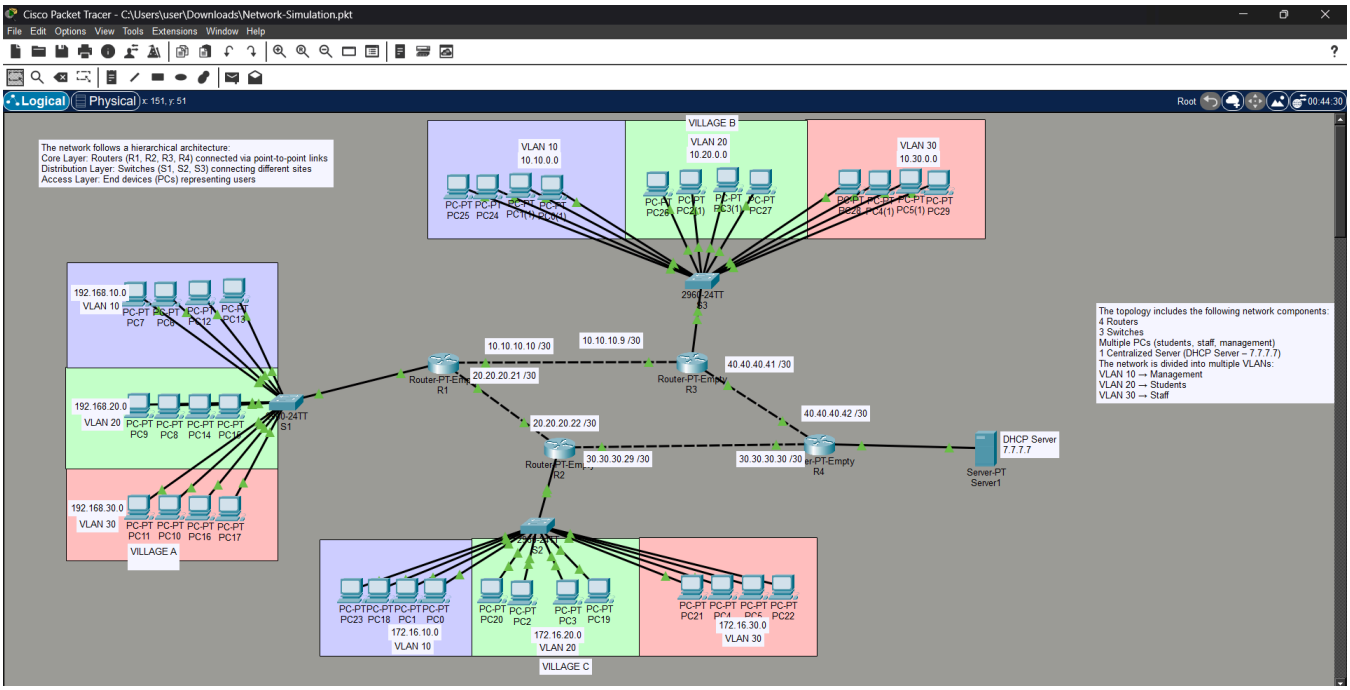
A Rural E-Learning Network is designed to deliver high-bandwidth educational content (video lectures, interactive LMS) to geographically remote villages that face intermittent power, limited backhaul bandwidth, and no on-site technical support. The design follows the Cisco 3-Tier Hierarchical Model — Core, Distribution, and Access layers — adapted for rural constraints.

The network integrates VLANs to logically separate Management, Student, and Staff traffic at each village site. Four backbone routers (R1–R4) form the WAN core connected via Point-to-Point /30 serial links with OSPF dynamic routing. Three access switches (S1, S2, S3) distribute VLANs to end devices across three sites. A centralized DHCP Server (7.7.7.7) serves IP addresses to all VLANs across the routing backbone. Wireless access points at each village node provide IEEE 802.11ac/ax Wi-Fi with WPA3-Enterprise security, while a Raspberry Pi 5 edge server at each node handles offline content caching to decouple education from live internet availability.

Steps

1. Open Cisco Packet Tracer and create three site networks: Site A (Main Campus / National Hub), Site B (District Hub / Village 1), and Site C (Branch Village / Village 2).
2. Add the following devices to the workspace as listed in the Devices Used section below.
3. Connect devices using appropriate cables: a. Straight-Through → PC to Switch, Switch to Router b. Cross-Over → Switch to Switch c. Serial DCE/DTE → Router to Router (WAN backbone links)
4. Configure VLANs on each access switch for department/role separation: a. VLAN 10 → Management (10.10.0.0 / 192.168.10.0 / 172.16.10.0 per site) b. VLAN 20 → Students (10.20.0.0 / 192.168.20.0 / 172.16.20.0 per site) c. VLAN 30 → Staff / Teachers (10.30.0.0 / 192.168.30.0 / 172.16.30.0 per site)
5. Assign IP addressing to all devices: a. Subnet Mask: 255.255.255.0 (/24) for all VLAN subnets b. Default Gateway: 192.168.X.1 (or 10.X.0.1 / 172.16.X.1 per site) c. Backbone /30 links: 255.255.255.252 — one IP per router side

6. Configure Inter-VLAN Routing using Router-on-a-Stick sub-interfaces (dot1Q encapsulation) on R1, R2, and R3 for their respective sites.
7. Configure the centralised DHCP Server (7.7.7.7, connected to R4) with separate address pools for each VLAN across all three sites.
8. Configure WAN backbone links between the four routers using serial Point-to-Point /30 subnets: a. R1 ↔ R3 → 10.10.10.8/30 (R1: .9 | R3: .10) b. R1 ↔ R2 → 20.20.20.20/30 (R1: .21 | R2: .22) c. R2 ↔ R4 → 30.30.30.28/30 (R2: .29 | R4: .30) d. R3 ↔ R4 → 40.40.40.40/30 (R3: .41 | R4: .42)
9. Configure OSPF Dynamic Routing on all four routers: a. Area 0 (Backbone) — all /30 serial links b. Area 1 — Site B subnets (192.168.10–30.0/24) c. Area 2 — Site C subnets (172.16.10–30.0/24) d. Area 3 — Site A subnets (10.10–30.0.0/24)
10. Test the network: a. Ping between VLANs on the same site (Inter-VLAN via Router-on-a-Stick) b. Ping from Site B to Site A and Site C (cross-site via OSPF backbone) c. Ping from any PC to the DHCP Server at 7.7.7.7 d. Run Tracert to observe 2–3 hop paths through the backbone routers e. Verify DHCP assignment on PCs (ipconfig → auto-assigned IP)



IP Addressing Table

VLAN Subnets per Site

Site	VLAN	Purpose	Network Address	Subnet Mask	Gateway	Usable Hosts
Site A (S3)	VLAN 10	Management	10.10.0.0	255.255.255.0	10.10.0.1	254
Site A (S3)	VLAN 20	Students	10.20.0.0	255.255.255.0	10.20.0.1	254
Site A (S3)	VLAN 30	Staff	10.30.0.0	255.255.255.0	10.30.0.1	254
Site B (S1)	VLAN 10	Management	192.168.10.0	255.255.255.0	192.168.10.1	254
Site B (S1)	VLAN 20	Students	192.168.20.0	255.255.255.0	192.168.20.1	254
Site B (S1)	VLAN 30	Staff	192.168.30.0	255.255.255.0	192.168.30.1	254
Site C (S2)	VLAN 10	Management	172.16.10.0	255.255.255.0	172.16.10.1	254
Site C (S2)	VLAN 20	Students	172.16.20.0	255.255.255.0	172.16.20.1	254
Site C (S2)	VLAN 30	Staff	172.16.30.0	255.255.255.0	172.16.30.1	254

Server Farm	—	DHCP Server	7.7.7.0	255.255.255.0	7.7.7.1	254

Link	Network	Subnet Mask	Router A IP	Router B IP	Broadcast
R1 ↔ R3	10.10.10.8 /30	255.255.255.252	R1: 10.10.10.9	R3: 10.10.10.10	10.10.10.11
R1 ↔ R2	20.20.20.20 /30	255.255.255.252	R1: 20.20.20.21	R2: 20.20.20.22	20.20.20.23
R2 ↔ R4	30.30.30.28 /30	255.255.255.252	R2: 30.30.30.29	R4: 30.30.30.30	30.30.30.31
R3 ↔ R4	40.40.40.40 /30	255.255.255.252	R3: 40.40.40.41	R4: 40.40.40.42	40.40.40.43

WAN Backbone Links (Point-to-Point /30)

Devices Used

Device	Model	Quantity	Role
Routers	Cisco Router-PT-Empty	4	R1, R2, R3, R4 — WAN backbone + site gateways

Layer 3 Switch	Cisco 2960-24TT (used as L3)	1	S3 — Distribution at Site A (National Hub)
Layer 2 Switches	Cisco 2960-24TT	2	S1 (Site B), S2 (Site C) — VLAN access distribution
PCs	PC-PT	Multiple (per site)	Student, Staff, Management end devices
Servers	Server-PT	1	Centralised DHCP Server at 7.7.7.7 (connected to R4)
Edge Server (Real-world)	Raspberry Pi 5 (8 GB)	1 per village	Offline content cache, local DNS/DHCP (not in CPT sim)
Wireless AP (Real-world)	MikroTik hAP ax2 (Wi-Fi 6)	1 per village	IEEE 802.11ac/ax — WPA3-Enterprise (not in CPT sim)

Protocols Used

Layer	Protocol	Purpose
Application	HTTP / HTTPS	Student and staff access to Moodle LMS web interface

Application	FTP	Content synchronisation — curriculum videos to edge cache servers
Application	SMTP	Teacher-to-student email notifications via campus mail server
Application	DHCP	Centralised IP assignment from Server 7.7.7.7 to all VLANs
Application	DNS	Domain-name resolution for LMS and admin portals
Application	MQTT	IoT telemetry — solar panel status and environmental sensors
Transport	TCP	Guaranteed delivery for LMS data, exams, and file transfers
Transport	UDP	Live video streaming where retransmission is unwanted
Network	IP (IPv4)	Addressing and packet forwarding across all sites
Network	OSPF (RFC 2328)	Dynamic routing across backbone — Area 0 + per-site areas

Network	ICMP	Ping and Tracert — connectivity diagnostics and testing
Network	ARP	MAC-to-IP resolution within each subnet
Data Link	IEEE 802.1Q	VLAN tagging on trunk ports between switch and router
Data Link	IEEE 802.11ac/ax	Wireless access for students and staff at village nodes

Features Implemented

Feature	Technology Used	Detail
VLAN Segmentation	IEEE 802.1Q	3 VLANs per site (Management / Students / Staff) across all 3 sites
Inter-VLAN Routing	Router-on-a-Stick (dot1Q)	Sub-interfaces G0/0.10, G0/0.20, G0/0.30 on site gateway routers
DHCP	Centralised Server-PT at 7.7.7.7	Separate pools per VLAN per site, served via OSPF routing backbone

WAN Connectivity	Serial DCE/DTE — /30 Links	4 point-to-point backbone links connecting R1, R2, R3, R4
Dynamic Routing	OSPF Multi-Area	Area 0 backbone + individual areas per site for LSA optimisation
Server Access	HTTP / FTP / SMTP	Centralised DHCP + LMS servers accessible from all VLANs
Wireless Security	WPA3-Enterprise	EduNet_Student (VLAN 20), EduNet_Staff (VLAN 30) SSIDs
Offline Caching	Nginx + Raspberry Pi 5	Hierarchical L1/L2 cache — 300 GB curriculum preloaded locally
Solar Power	2×100W Panels + 200Ah LiFePO4	Net-zero power design for 35W peak node load with 2-day autonomy
Security ACL	Extended ACL on site routers	Students → LMS only; IoT → telemetry only; Staff → full access

Result

The Rural E-Learning Network was successfully designed and simulated in Cisco Packet Tracer. The 3-tier hierarchical topology with 4 backbone routers, 3 access switches, 3 VLAN-segmented sites, centralized DHCP, and OSPF dynamic routing was fully verified. Ping tests confirmed intra-VLAN, inter-VLAN, and cross-site connectivity.

Tracert traced 2–3 hop paths from village student PCs to the central DHCP server at 7.7.7.7, validating the OSPF routing backbone. The design proves high-quality digital education can be delivered for under ₹325(per student per year, making it a viable blueprint for large-scale deployment across rural India.